AR Parameter estimation

$$y[t] = -a_1 y[t-1] - a_2 y[t-2] - \dots - a_p y[t-p] + f[t]$$

when $f[t]$ is
- zero-mean
- white
- variance σ_f^2

AR(p)

autoregressive

1. Mult.
2. Take expectations

$$E(y[t]\, y[t-\ell]) = E[-a_1 y[t-1] y[t-\ell] - a_2 y[t-2] y[t-\ell] - \dots - a_p y[t-p] y[t-\ell] + f[t]\, y[t-\ell]]$$

$$r_y[\ell] = -a_1 r_y[\ell-1] - a_2 r_y[\ell-2] - \dots - a_p r_y[\ell-p] + E[f[t]\, y[t-\ell]]$$

Expectations

Ex: X: throw a die

$P(X=1) = \frac{1}{6}$

$P(X=2) = \frac{1}{6}$

$\vdots$

$P(X=6) = \frac{1}{6}$

$$E[X] = 1 \cdot P(X=1) + 2\, P(X=2) + \dots + 6\, P(X=6)$$

$$= \frac{1}{6}(1+2+3+4+5+6)$$

$$= \underline{3.5}$$

Ex: X = swap takes

$P(X = 10^6)$

$P(X = 0)$

$$E[X] = 10^6 \cdot \underbrace{P(X=10^6)}_{10^{-7}} + 0 \cdot \underbrace{P(X=0)}_{1-10^{-7}}$$

$$= 10^{-1}$$

Discrete $\quad E[X] = \sum_x x\, P(X=x)$

Cont's $\quad E[X] = \int_{-\infty}^{\infty} x\, P_X(x)\, dx$

- Expectations are linear operators.

$$E[a_1 X_1 + a_2 X_2 + \dots + a_n X_n] = E[a_1 X_1] + E[a_2 X_2] + \dots + E[a_n X_n]$$

- Non random factors can be pulled out of expectations.

$$= a_1 E[X_1] + a_2 E[X_2] + \dots + a_n E[X_n]$$

$y[n]$: random process.

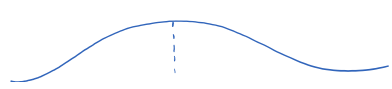
$E[y[n]]$ = mean.

Autocorrelation.

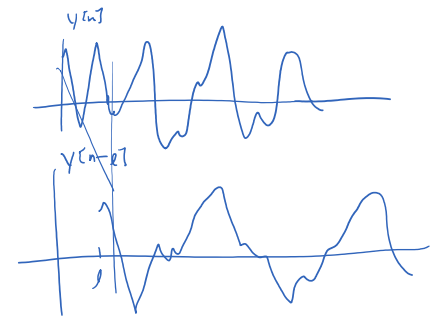
$y[n]$

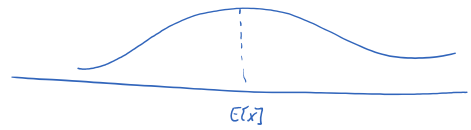
Mean: $E[X]$

Variance: $\sigma^2 = E[(X - E[X])^2] = E[X^2] - (E[X])^2$



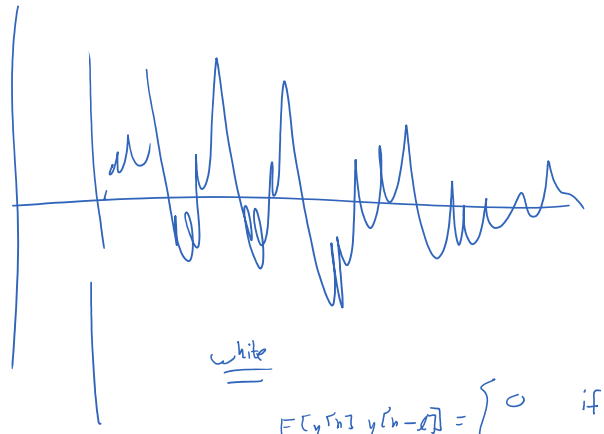
Autocorrelation.





$r_y(l) = r(l) = E[\, y[n]\, y[n-l] \,]$

$r_{yy}(l)$



white

$$E[y[n]\, y[n-l]] = \begin{cases} 0 & \text{if } l \neq 0 \\ \sigma_y^2 & \text{if } l = 0 \end{cases}$$

$$\boxed{r_y[l] = -a_1\, r_y[l-1] - \cdots - a_p\, r_y[l-p]} + \underbrace{E[\, y[n-l]\, f[n] \,]}_{0 \text{ if } l>0}$$

$$\begin{array}{l} l=1 \\ l=2 \\ \\ l=p \end{array} \underbrace{\begin{bmatrix} r_y[0] & r_y[-1] & \cdots & r_y[1-p] \\ r_y[1] & r_y[0] & \cdots & r_y[2-p] \\ \vdots & & & \\ r_y[p-1] & r_y[p-2] & \cdots & r_y[0] \end{bmatrix}}_{R} \underbrace{\begin{bmatrix} -a_1 \\ -a_2 \\ \vdots \\ -a_p \end{bmatrix}}_{(\,q\,)} = \underbrace{\begin{bmatrix} r_y[1] \\ r_y[2] \\ \vdots \\ r_y[p] \end{bmatrix}}_{P}$$

Yule-Walker

Toeplitz matrix

Real: $R = R^T$ (symmetric)

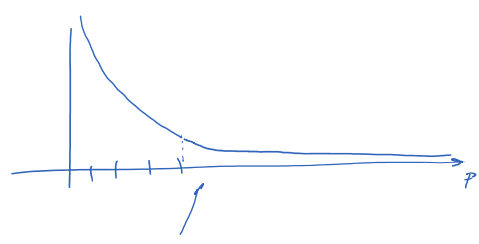
Complex: $R = R^H$

H: conjugate transpose.

$R\, q = P.$ $\qquad q = R^{-1} P.$

$$\begin{bmatrix} r[0] \end{bmatrix}$$

$$\begin{bmatrix} \boxed{r[0]} & r[-1] \\ r[1] & r[0] \end{bmatrix}$$

$$\begin{bmatrix} \boxed{\begin{matrix} r[0] & r[-1] \\ r[1] & r[0] \end{matrix}} & \begin{matrix} r[-2] \\ r[-1] \end{matrix} \\ \begin{matrix} r[-2] & r[-1] \end{matrix} & r[0] \end{bmatrix}$$

$E[(y[n] - \hat{y}_p[n])^2]$